The Relational Model

Topics List

- Relational Model Terminology
- Properties of Relations
- Keys
- Integrity Constraints

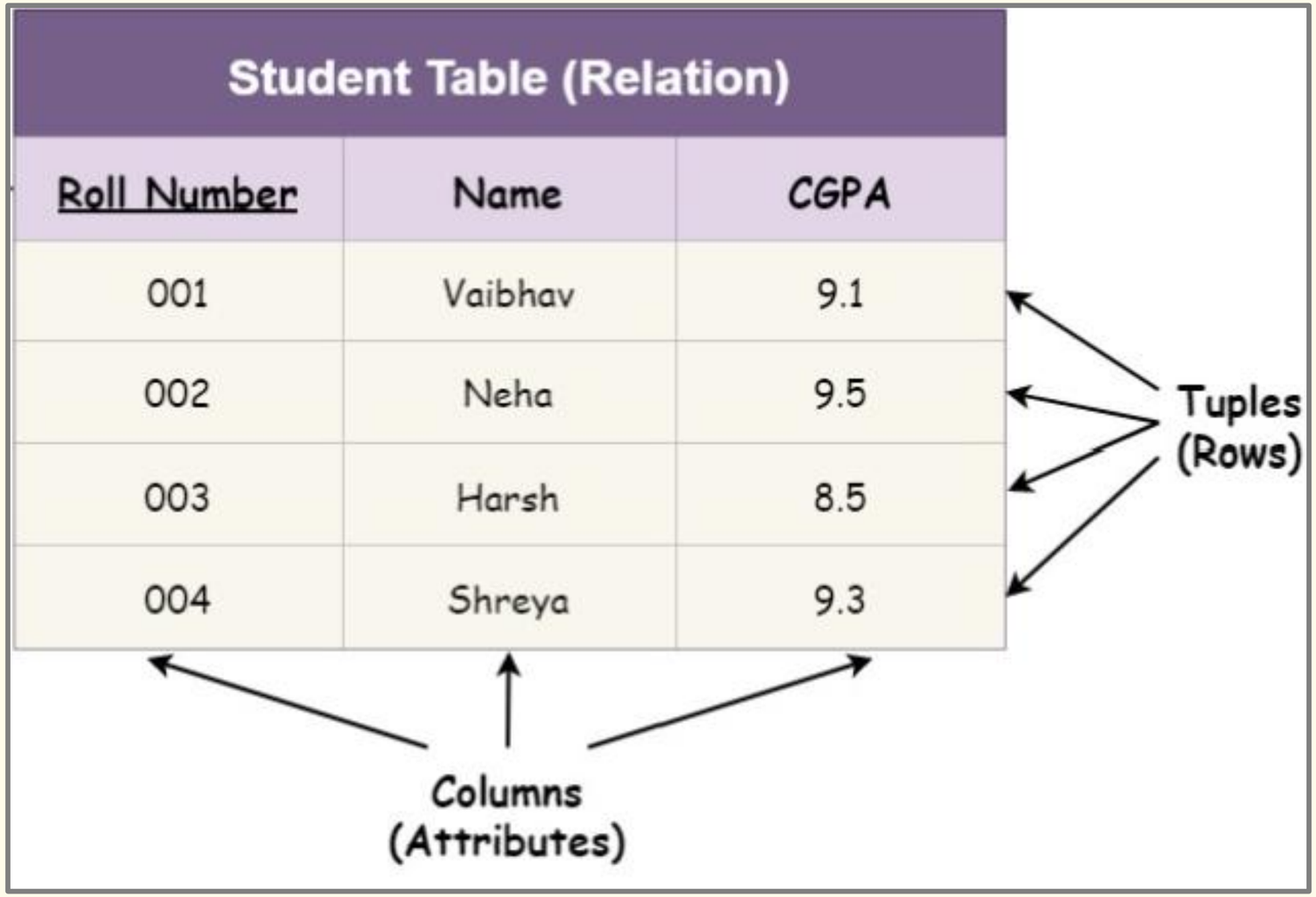
Relational Model Terminology

- **A relation** is a table with columns and rows.
  - Only applies to logical structure of the database, not the physical structure.
- **Attribute** is a named column of a relation.
- **Domain** is the set of allowable values for one or more attributes.

Relational Model Terminology

- **Tuple** is a row of a relation.
- **Degree** is the number of attributes in a relation.
- **Cardinality** is the number of tuples in a relation.
- **Relational Database** is a collection of normalised relations with distinct relation names.

Relational Model Terminology



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Properties of Relations

- Relation name is distinct from all other relation names in relational schema.
- Each cell of relation contains exactly one atomic (single) value.
- Each attribute has a distinct name.
- Values of an attribute are all from the same domain.

Properties of Relations

- Each tuple is distinct; there are no duplicate tuples.
- Order of attributes has no significance.
- Order of tuples has no significance, theoretically.

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Keys

- A database is used to store data for retrieval. An attribute that may be used to find a particular entity occurrence is called a **key**.
- A **key** is an attribute or collection of attributes that uniquely identifies an entity occurrence.
- For example, the *staffNo* value for each staff member makes him/her identifiable among staff.

Relational Keys

- Candidate Key
  - Candidate keys are defined as the minimal set of fields which can uniquely identify each record/tuple in a table. There can be more than one candidate key.
- Primary Key
  - Candidate key selected to identify records uniquely within a relation.
- Alternate Keys
  - Candidate keys that are not selected to be primary key.
- Foreign Key
  - Attribute, or set of attributes, within one relation that matches the primary key of another relation.

Relational Keys

Primary Key

Referenced/Primary Table

Table 1 : department_info

dept_ID	dept_name	dept_location	dept_budget
101	IT	FC-89	500000
102	Electronics	FC-54	400000
103	Admin	TY-56	350000
104	Accounts	TY-43	250000

Foreign Key

Referencing/Related Table

Table 2 : employee_info

emp_ID	emp_name	emp_dob	emp_city	dept_id
SL21	Sana	30-Apr-1978	Delhi	101
SG45	Ram	08-Jan-1987	Prayagraj	103
PL67	Aayush	01-Mar-1992	Delhi	101
HG56	Dawid	25-Jul-1990	Kanpur	103
NF78	Nishank	19-Nov-1996	Prayagraj	104
FR32	John	03-Aug-1989	Delhi	104

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Integrity Constraints

- Null
  - Represents value for an attribute that is currently unknown or not applicable for this tuple.
  - Deals with incomplete or exceptional data.
  - Represents the absence of a value and is not the same as zero or spaces, which are values.

Integrity Constraints

- Domain constraints
  - Domain constraints can be defined as the definition of a valid set of values for an attribute.
  - The data type of domain includes string, character, integer, time, date, currency, etc. The value of the attribute must be available in the corresponding domain

ID	NAME	SEMENSTER	AGE
1000	Tom	1st	17
1001	Johnson	2nd	24
1002	Leonardo	5th	21
1003	Kate	3rd	19
1004	Morgan	8th	A

Not allowed. Because AGE is an integer attribute

Integrity Constraints

- Entity Integrity constraint
  - In a base relation, no attribute of a primary key can be null.
  - Primary key must also be unique.

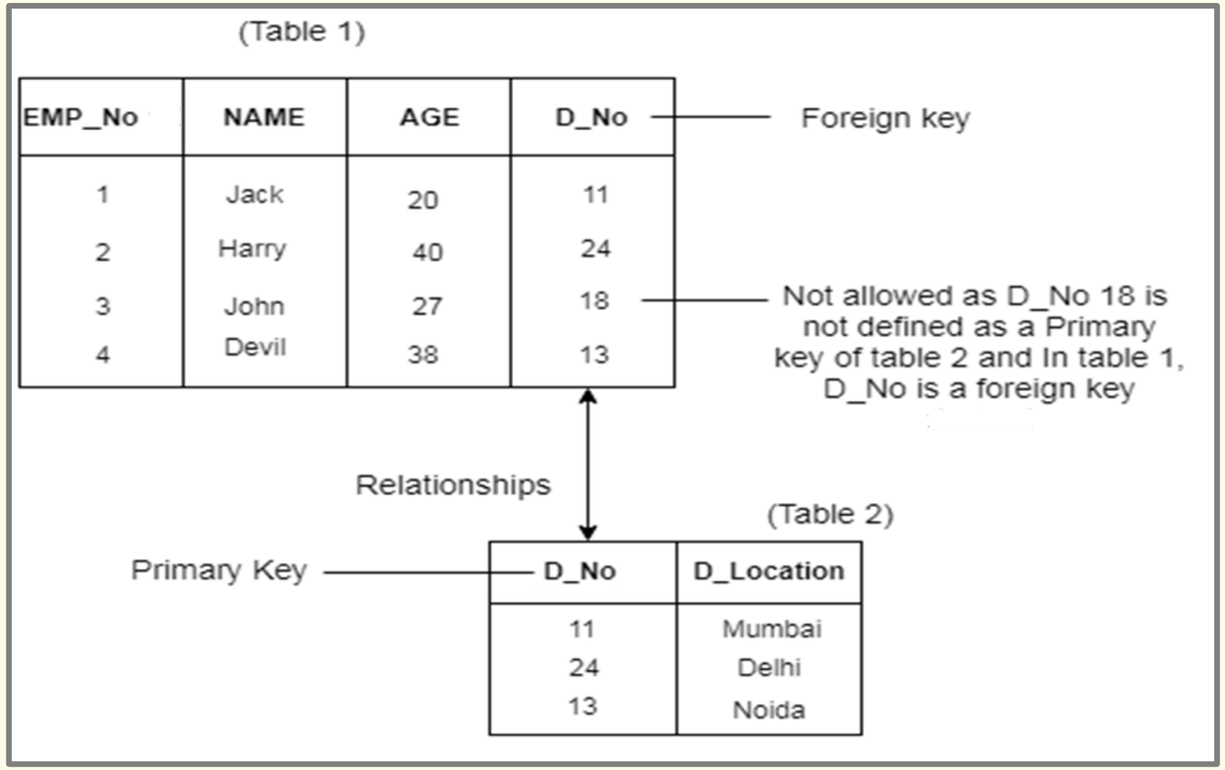
EMPLOYEE

EMP_ID	EMP_NAME	SALARY
123	Jack	30000
142	Harry	60000
164	John	20000
	Jackson	27000

Not allowed as primary key can't contain a NULL value

Integrity Constraints

- **Referential Integrity**
  - If foreign key exists in a relation (Table 1), either the foreign key value must match the primary (or alternate) key value of some tuple in its home relation (Table 2) or the foreign key value must be wholly null.



Integrity Constraints

- General Constraints
  - Additional rules specified by users or database administrators that define or constrain some aspect of the enterprise.
  - For Example:
    In the PropertyForRent table, type must be House, Flat or Apartment.

Sample Relations

- These 2 relations adhere to the properties of a relation:
  - Relation names are distinct.
  - Each cell of each relation contains one value.
  - Each attribute has a distinct name.
  - Values of an attribute are all from the same domain (e.g. *city* column, *position* column).
  - Each tuple is distinct; there are no duplicate tuples because of the primary key.
- These relations also adhere to the Entity and Referential Integrity constraints.

Branch

branchNo	street	city	state	zipCode	telNo
B001	8 Jefferson Way	Portland	OR	97201	503-555-3618
B002	City Center Plaza	Seattle	WA	98122	206-555-6756
B003	14 – 8th Avenue	New York	NY	10012	212-371-3000
B004	16 – 14th Avenue	Seattle	WA	98128	206-555-3131

Primary key branchNo

Staff

staffNo	firstName	lastName	position	salary	branchNo
S1500	Tom	Daniels	Manager	46000	B001
S0003	Sally	Adams	Assistant	30000	B001
S0010	Mary	Martinez	Manager	50000	B002
S3250	Robert	Chin	Supervisor	32000	B002
S2250	Sally	Stern	Manager	48000	B004
S0415	Art	Peters	Manager	41000	B003

Primary key staffNo

Foreign key branchNo references Branch(branchNo)